MOMENTUM STRATEGIES IN A MODERN U.S. EQUITY SAMPLE
Replication of Jegadeesh and Titman (1993)

1. Research question
Do stocks with strong past returns continue to outperform stocks with weak past returns? This project applies a market-model approach to monthly momentum portfolios and examines whether winner-minus-loser returns vary with firm size.

2. Data and sample
The current analysis covers December 2001 through December 2025, a modern sample period used in the course brief. The analysis-ready inputs contain monthly returns for decile portfolios P1-P10, the winner-minus-loser spread (WML), a market return, and the monthly risk-free rate. Size-sorted results are available for small, medium, and large firms.

This is a modern-sample application of the paper's method, not an exact numerical replication of the original historical sample. Licensed course/CRSP source data are not published in this repository. Users must obtain authorized inputs through the course and construct the documented local analysis files.

3. Empirical design
For each decile portfolio, estimate:
$R_{p,t} - RF_t = \alpha_p + \beta_p (R_{m,t} - RF_t) + \varepsilon_{p,t}$.
For WML, regress the winner-minus-loser spread on market excess return. Do not subtract RF from WML a second time because WML is already a long-short spread. Use Newey-West/HAC standard errors with five lags.

4. Core replication results
The currently checked Stata estimates show a full-sample WML alpha of 0.011317 per month (about 1.132 percent per month), with a HAC t-statistic of 2.94. This is statistically significant at the 1 percent level using conventional two-sided critical values and indicates a positive market-adjusted momentum spread.

The loser portfolio P1 has alpha -0.01035 per month (t = -2.22), significant at the 5 percent level. The winner portfolio P10 has alpha 0.000966 (t = 0.44), which is not statistically significant. The full-sample pattern is therefore more clearly associated with a negative alpha for losers than a statistically significant positive alpha for winners.

5. Results by firm size
The WML market-model estimates by size group are:

Sample	Monthly alpha	HAC t-statistic
Small	0.014671 (1.467%)	3.11
Medium	0.010555 (1.056%)	3.04
Large	0.007460 (0.746%)	2.65

All three estimates are positive and statistically significant at approximately the 1 percent level. The point estimate is largest among small firms and declines across medium- and large-size groups. This pattern is consistent with stronger momentum among smaller firms, but the ordering of point estimates alone does not show that the differences are statistically significant.

6. Independent extension: size heterogeneity
The extension formally tests pairwise differences in size-group WML alphas. For each pair, estimate the intercept in a time-series regression of the difference between the two groups' WML returns on market excess return, using HAC standard errors with five lags. The intercept estimates the difference in their market-model alphas under a common time index.

The extension output must be generated by running code/run_all.py with the approved local input files. Its pairwise estimates and t-statistics have not yet been checked against those files, so no conclusion about statistically significant cross-size differences is made in this draft.

7. Comparison with the original paper
The positive WML spread is qualitatively consistent with the broad momentum finding of Jegadeesh and Titman (1993). Exact numerical equality is not expected because this project uses a modern 2001-2025 sample rather than the original paper's historical period. Data coverage and implementation choices may also differ.

8. Validation, limitations, and conclusion
Before submission, verify monthly date alignment, return units, missing observations, the market and risk-free-rate merges, and the construction WML = P10 - P1. Compare Python outputs with independent Stata estimates. The reported core estimates provide evidence of a positive market-adjusted momentum spread in the modern sample, with positive point estimates in all size groups. The formal extension must be run before concluding whether size-group alphas differ statistically.

References
Jegadeesh, N., and S. Titman (1993). Returns to Buying Winners and Selling Losers: Implications for Stock Market Efficiency. The Journal of Finance, 48(1), 65-91. DOI: 10.1111/j.1540-6261.1993.tb04702.x

Draft status: this PDF summarizes currently checked Stata core results. Update it after the Python workflow has been run on the approved inputs and all extension results and figures have been independently validated.